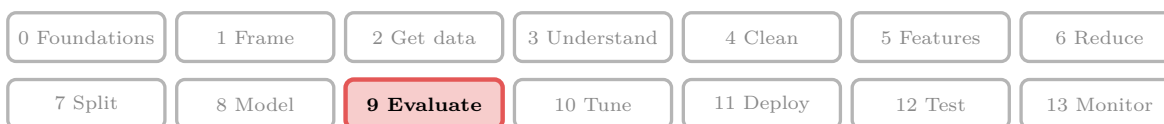


Regression Metrics: MAE, MSE, RMSE, R^2 and Adjusted R^2

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Where this fits: Pipeline map step 9 (Evaluate). See the [Course map](#).



- **Builds on:** Best-fit line and squared error ([Note 51](#)).
- **Leads to:** Regression trees ([Note 99](#)); ANN for regression ([Note 1013](#)).
- **Compare with:** Loss functions in deep learning ([Note 1014](#)).

1. Overview

Key point: A regression metric turns all the errors on the test set into one number. Five are common: MAE, MSE, RMSE, R^2 and adjusted R^2 .

After training a regression model, we need to know how good it is. A **regression metric** (G-1652) compares the model's predictions with the true values on the test set and summarises the errors as one number, like a report card that sums up a term's work in one grade.

Three words recur. A **feature** (G-772) is an input variable (one column of the data table), here CGPA. The **target** (G-1949) is the output we predict, here the package. An **observation** (G-1374) is one record (one row), here one student.

This Note covers five metrics, in order:

Metric	Answers
MAE	How far off is a typical prediction, in the target's units?
MSE	The average squared error; what training minimises
RMSE	MSE brought back to the target's units
R^2	How much better than guessing the average is the model?
Adjusted R^2	R^2 , but with a penalty for useless features

No single metric is best for every problem, which is why several exist. All the numbers below come from the simple linear regression of the previous Notes: CGPA to package, tested on 40 students.

Throughout, y_i is a student's actual package, $\hat{y}_i$ the model's prediction, and n the number of test students.

Figure 1 shows what every metric starts from. Each test student has an error: the vertical stick between the actual package and the line's prediction. Each metric below is a different way to turn these 40 sticks into one number.

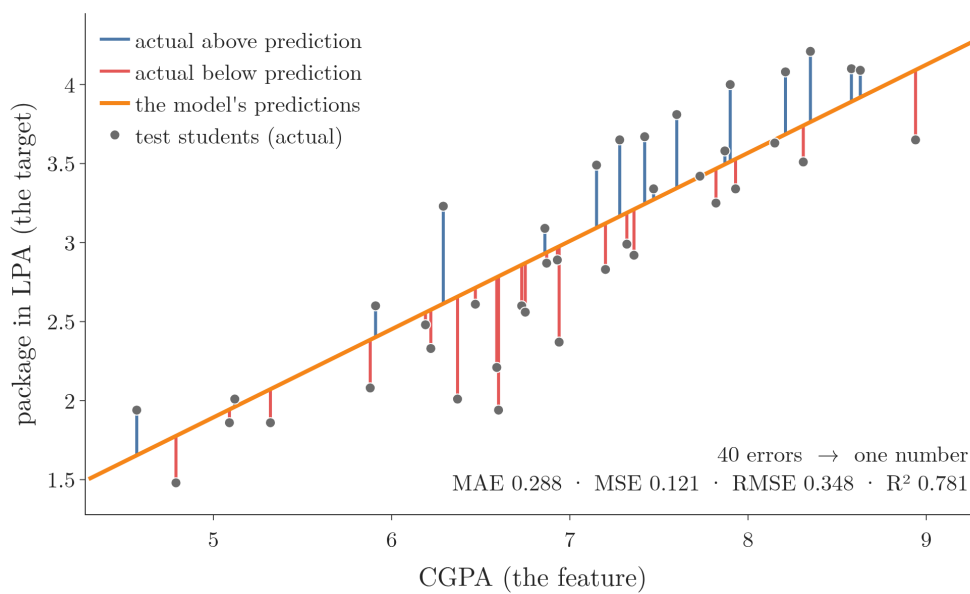


Figure 1: The 40 test students, the regression line (the model's predictions) and each student's error as a stick: blue when the actual package is above the prediction, red when below. The five metrics summarise these 40 errors.

2. Mean absolute error (MAE)

Key point: MAE is the average size of the errors, ignoring their sign. On the placement data it is 0.29 LPA.

The plain idea: measure how long each error stick of Figure 1 is, ignore whether it points up or down, and average the lengths. This average is the **mean absolute error (MAE)** (G-1194). Figure 2 does it step by step on the 40 test students.

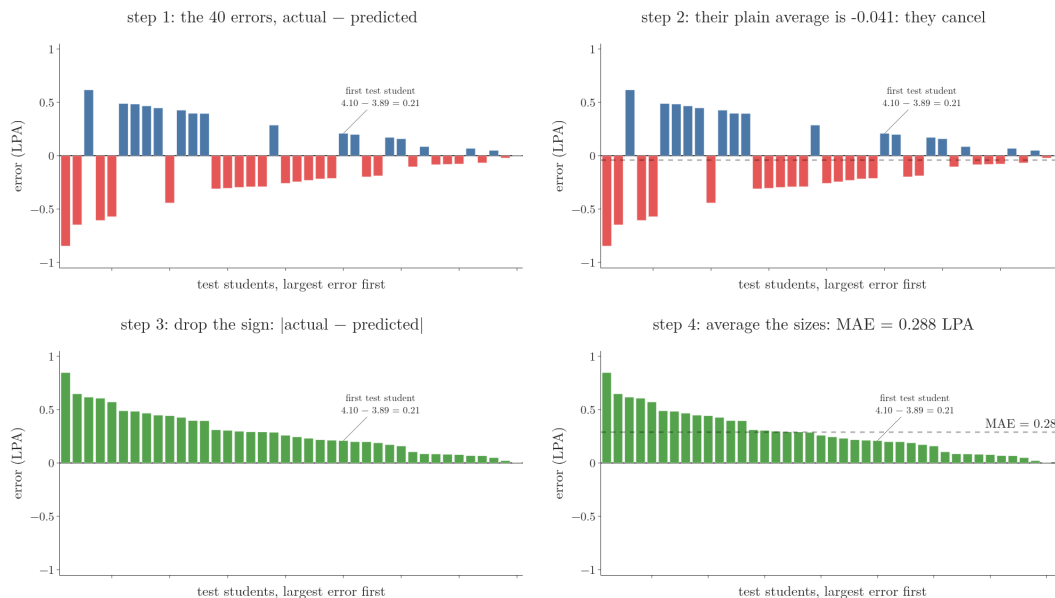


Figure 2: MAE in four steps. 1: the 40 errors with their signs. 2: their plain average is -0.041 , because positive and negative errors cancel. 3: drop the sign. 4: the average size is $\text{MAE} = 0.288$ LPA. The labelled bar is the first test student, $4.10 - 3.89 = 0.21$.

In words: for each point, take the gap between the actual and the predicted value, drop its sign, and average these gaps.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

With numbers: the first test student has an actual package of 4.10 and a prediction of 3.89, an absolute error of 0.21. Averaging all 40 such errors gives

$$\text{MAE} = 0.288 \text{ LPA}$$

The sign is dropped because some points lie above the line and some below; without the absolute value, positive and negative errors would cancel. Step 2 of Figure 2 shows how badly: the signed errors average to -0.041 , as if the model were almost perfect.

Advantages:

- **Same units as the target.** The model is off by about 0.29 lakh rupees per year on average, a statement anyone can understand.
- **Robust to outliers.** An **outlier** (G-1420) is an observation far from the rest. One very wrong prediction raises MAE only in proportion to its size (Section 5).

Disadvantage: the absolute value has a sharp corner at 0, so it cannot be differentiated there. The corner makes MAE awkward to use as the function that training minimises (the previous Note chose squares for exactly this reason).

3. Mean squared error (MSE)

Key point: MSE averages the squared errors. MSE is smooth and punishes large errors heavily, but its units are squared. On the placement data it is 0.121.

In words: square each error and average the squares. The result is the **mean squared error (MSE)** (G-1201).

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

With numbers: the first student's error of 0.21 becomes $0.21^2 = 0.044$. Averaging all 40 squares gives

$$\text{MSE} = 0.121$$

Advantage: the square is smooth everywhere, so MSE can be differentiated. Smoothness is why MSE is used as the **loss function** (G-1130) that models minimise during training, as in the previous Note.

Disadvantages:

- **Squared units.** 0.121 is in LPA^2 , which has no everyday meaning.
- **Sensitive to outliers.** Squaring makes a large error dominate: an error of 3 counts 9, while ten errors of 0.3 together count only 0.9.

Figure 3 shows the effect on the 40 test errors. Watch the right panel: squaring flattens the small errors and stretches the large ones.

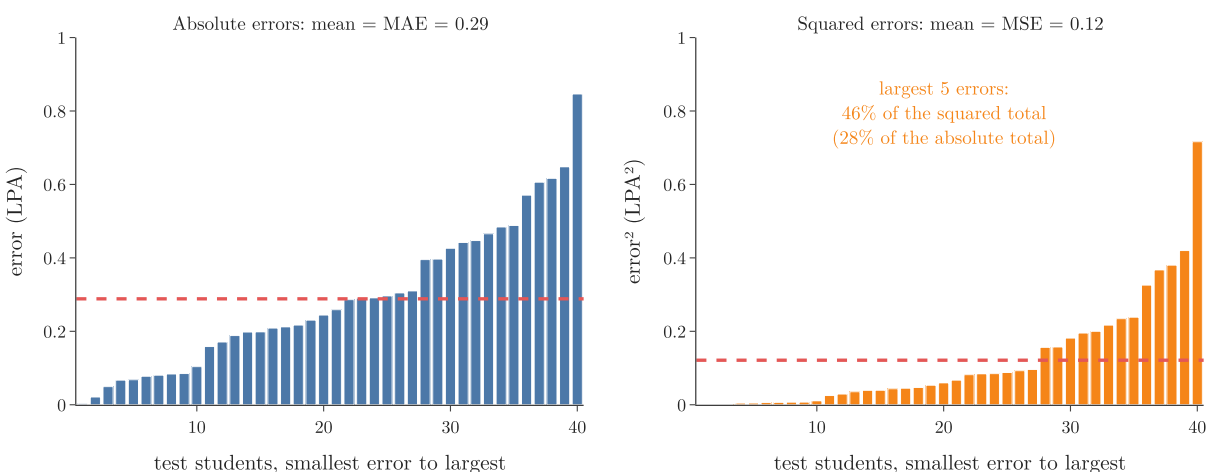


Figure 3: The 40 test errors sorted by size: absolute (left) and squared (right). The dashed line is the mean, MAE on the left and MSE on the right.

The 5 largest errors make up 28% of the absolute total but 46% of the squared total.

4. Root mean squared error (RMSE)

Key point: RMSE is the square root of MSE: back in the target's units, but still punishing large errors. On the placement data it is 0.35 LPA.

In words: compute the MSE, then take its square root. The result is the **root mean squared error (RMSE)** (G-1705).

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

With numbers:

$$\text{RMSE} = \sqrt{0.121} = 0.348 \text{ LPA}$$

RMSE is in LPA again, like MAE, so it is easy to report. RMSE is always at least as large as MAE (here 0.35 against 0.29), because the squares give extra weight to the larger errors. The two are equal only when every error has the same size.

Figure 4 makes both numbers into lengths. Draw each test error as a square with side equal to the error's size (grey). MSE is the average area of these squares, 0.121. The orange square has exactly that area, so its side is RMSE, 0.348. MAE is the average side, 0.288 (blue). The few big squares, such as the one with side 0.85, add a lot of area, so the square with the average area is longer than the average side.

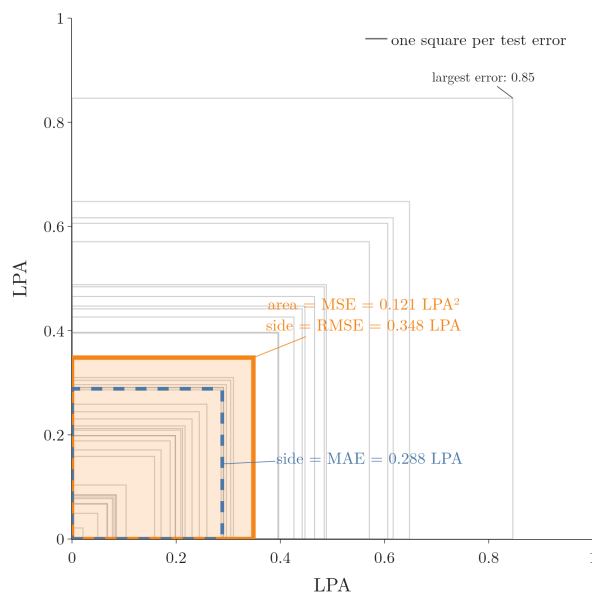


Figure 4: Each of the 40 test errors drawn as a square from the same corner (grey). Orange: the square whose area is the average area (MSE = 0.121 LPA squared); its side is RMSE = 0.348 LPA. Dashed blue: the square whose side is the average side, MAE = 0.288 LPA.

Extra: The proof. Write $a_i = |y_i - \hat{y}_i|$, so MAE is the average of the a_i . Then

$$\text{MSE} - \text{MAE}^2 = \frac{1}{n} \sum a_i^2 - \text{MAE}^2 = \frac{1}{n} \sum (a_i - \text{MAE})^2 \geq 0$$

so $\text{RMSE} = \sqrt{\text{MSE}} \geq \text{MAE}$. The gap is the spread of the error sizes.

Python: The three error metrics.

```
from sklearn.metrics import (mean_absolute_error,
                             mean_squared_error, root_mean_squared_error)
```

```
mean_absolute_error(y_test, y_pred)      # 0.288
```

```
mean_squared_error(y_test, y_pred)      # 0.121
```

```
root_mean_squared_error(y_test, y_pred) # 0.348
```

Older code computes RMSE as `np.sqrt(mean_squared_error(...))`; both give the same number. Older scikit-learn also had `mean_squared_error(..., squared=False)`, which has since been removed.

5. MAE versus RMSE with an outlier

Key point: One very wrong prediction barely moves MAE but pushes RMSE up sharply.

Figure 5 takes the 40 test predictions and makes one of them worse and worse.

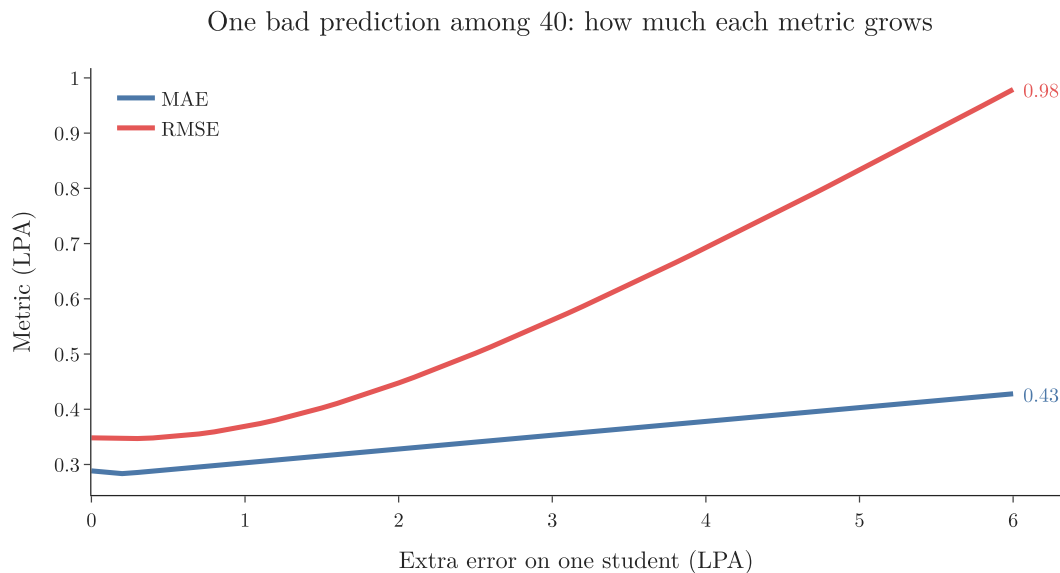


Figure 5: One bad prediction among 40: MAE and RMSE

With one prediction 6 LPA off, MAE rises from 0.29 to 0.43, while RMSE rises from 0.35 to 0.98. The squares let a single error dominate.

So the choice depends on the problem:

- If occasional large errors are what matters most (for example, a badly wrong price estimate is very costly), use RMSE: it makes them visible.
- If the data contains outliers that should not dominate the assessment, use MAE.

6. R^2 score

Key point: R^2 compares the model with the simplest possible model, “always predict the average”. R^2 is 1 for a perfect model and 0 for a model no better than the average. On the placement data it is 0.78.

6.1 The idea

Key point: R^2 measures how much of the error of “guessing the average” the model removes.

MAE, MSE and RMSE have a weakness: their size depends on the units. An RMSE of 0.35 is small for packages in LPA but would be huge for a target measured in thousands. We need a comparison point.

The simplest model of all ignores the feature and predicts the average package, 2.96 LPA, for every test student: the flat line in the first frame of Figure 6. Each error is drawn as a square, so the total orange area is the sum of squared errors: 22.13. Watch the line turn into the regression line: the squares shrink until their total is only 4.85, and the R^2 bar fills.

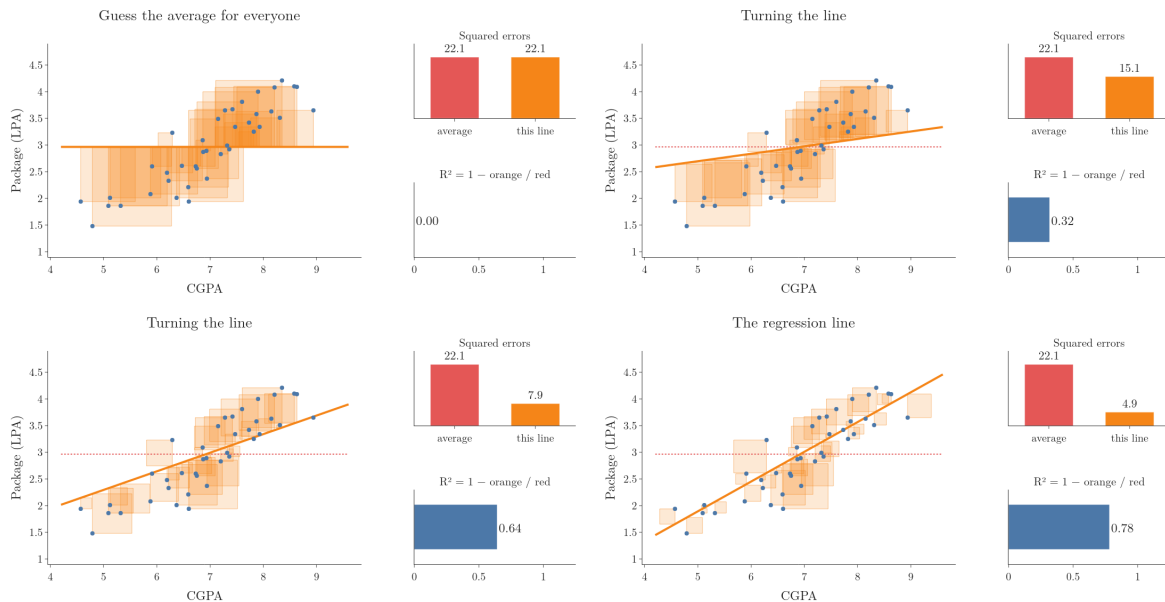


Figure 6: The average line turns into the regression line. Each square’s side is one student’s error; the bars show the total squared error and R^2 . Comparing the error around the average line with the error around the regression line follows StatQuest’s “R-squared, Clearly Explained!!!” (Starmer).

6.2 The formula

Key point: R^2 is 1 minus (squared errors of the model) divided by (squared errors of the average).

In words: divide the model’s sum of squared errors by the average-only model’s sum of squared errors, and subtract the result from 1.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Here SS_{res} (the **residual sum of squares**, G-1684) is the model’s total squared error, and SS_{tot} (the **total sum of squares**, G-1993) is the total squared error of always predicting the mean $\bar{y}$.

With numbers from Figure 6:

$$R^2 = 1 - \frac{4.85}{22.13} = 1 - 0.219 = 0.781$$

Python: R^2 score.

```
from sklearn.metrics import r2_score
r2_score(y_test, y_pred) # 0.781
```

6.3 Reading R^2

Key point: 0.78 means the model removes 78% of the error of guessing the average; the CGPA explains 78% of the variation in packages.

- $R^2 = 1$: every prediction is exact ($SS_{res} = 0$).
- $R^2 = 0$: the model is no better than predicting the average.
- $R^2 < 0$: the model is worse than predicting the average, which can happen on a test set with a bad model.

Whether R^2 can be negative depends on which data it is computed on:

- **On the training data** of a least-squares line, R^2 is always between 0 and 1. The flat average line is itself one of the candidate lines, and least squares picks the line with the smallest squared error, so the fitted line can never do worse than the average line: $SS_{res} \leq SS_{tot}$.
- **On test data** nothing guarantees this. The line was fitted to other observations, so it can miss the test observations by more than their own average does. Then $SS_{res} > SS_{tot}$ and R^2 is negative (scikit-learn `r2_score` docs).

Here, $R^2 = 0.78$: CGPA explains about 78% of the variation in packages; the remaining 22% comes from things not in the data, such as interviews (the stochastic errors of the earlier Note).

R^2 is also called the **coefficient of determination** (G-1717).

The name R^2 is no accident. For simple linear regression, R^2 on the training data equals the square of the correlation coefficient r between the feature and the target (see the [linear regression maths Note](#), section 5.1). For CGPA and package on the 160 training students, $r = 0.879$ and $r^2 = 0.879^2 = 0.773$, exactly the training R^2 in the table of Section 7.1.

Squaring makes correlations easier to compare. A correlation of 0.7 gives $R^2 = 0.49$ and a correlation of 0.5 gives $R^2 = 0.25$: the first feature explains about twice as much of the variation as the second, which the raw values 0.7 and 0.5 do not show.

7. Adjusted R^2

Key point: Adding features can push R^2 up even when the new features are useless. Adjusted R^2 subtracts a penalty for each feature, so it only rises when a feature truly helps.

7.1 The problem with R^2

Key point: On the training data, R^2 never goes down when a feature is added, even a feature of random numbers.

Suppose we add a feature of random numbers to the placement data, one that has nothing to do with packages. On the training data, R^2 still creeps up. The new feature gives the model

one more number to tune, and setting that number to 0 gives back the old model, so the fit on the training data can never get worse. A random feature almost always matches the leftover errors a little by chance, so R^2 usually rises.

Figure 7 adds 0 to 20 features of random numbers next to CGPA. The split is the same each time; only the random numbers change, and every score is the average over 200 draws of them.

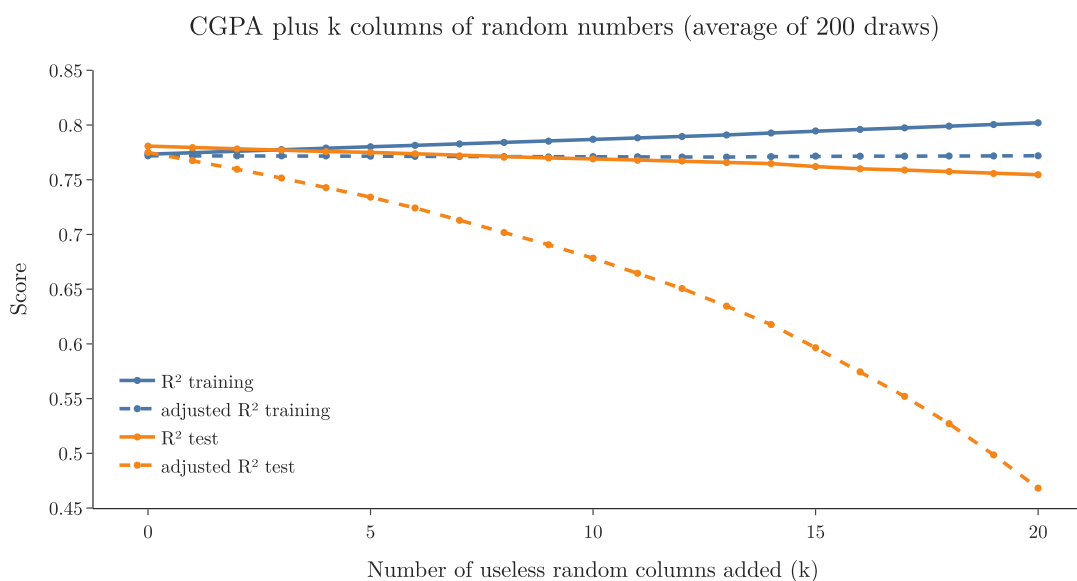


Figure 7: R^2 and adjusted R^2 as useless random features are added

Random features added	0	1	5	10	20
R^2 , training	0.773	0.775	0.780	0.787	0.802
Adjusted R^2 , training	0.772	0.772	0.772	0.771	0.772
R^2 , test	0.781	0.779	0.775	0.769	0.755
Adjusted R^2 , test	0.775	0.767	0.734	0.678	0.468

The training R^2 rises from 0.773 to 0.802, which looks like progress. The test R^2 shows the truth: it falls from 0.781 to 0.755, because the extra features are noise. Adjusted R^2 on the training data stays at 0.772, so it reports the true story without needing a test set.

7.2 The formula

Key point: Adjusted R^2 scales the unexplained part by $(n - 1)/(n - 1 - k)$, which grows with the number of features k .

In words: **adjusted R^2** (G-177) takes the part R^2 leaves unexplained, $1 - R^2$, makes it larger according to how many features k the model uses compared with the number of observations n , and subtracts that from 1 (ISLR §6.1.3).

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - 1 - k}$$

With numbers, for CGPA alone on the test set ($R^2 = 0.781$, $n = 40$, $k = 1$):

$$R_{adj}^2 = 1 - \frac{0.219 \times 39}{38} = 1 - 0.225 = 0.775$$

How it behaves:

- **A useless feature:** k goes up by 1, which makes the fraction larger, while R^2 barely changes. Adjusted R^2 falls.
- **A useful feature:** R^2 rises enough to outweigh the penalty. Adjusted R^2 rises.

In Figure 7, adjusted R^2 on the training data stays flat at 0.772 however many random features are added: adjusted R^2 is not fooled. On the small test set (40 observations), the penalty is strong, and adjusted R^2 falls from 0.775 to 0.468 with 20 random features.

Extra: A version of this demonstration sometimes builds a “useful” feature, such as an IQ score, by adding small noise to the package itself. Such a feature is made from the answer, so it would never exist in real data. It is **target leakage** (G-1948): information about the answer that the model should not have (Kaufman et al. 2012). The honest version of the lesson is the one above: useless features raise R^2 on the training data, and adjusted R^2 exposes them. Adjusted R^2 is most useful with multiple linear regression, the next Note.

Extra: scikit-learn has no function for adjusted R^2 ; we compute it from `r2_score` with the formula above. Here n is the number of observations in the set being scored (40 for the test set) and k the number of features.

8. Summary

Metric	Formula	Placement data	Units	Outliers
MAE	$\frac{1}{n} \sum y_i - \hat{y}_i $	0.288	LPA	robust
MSE	$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$	0.121	LPA ²	sensitive
RMSE	$\sqrt{\text{MSE}}$	0.348	LPA	sensitive
R ²	$1 - SS_{res}/SS_{tot}$	0.781	none	sensitive
Adjusted R ²	$1 - (1 - R^2) \frac{n-1}{n-1-k}$	0.775	none	sensitive

- MAE and RMSE are in the target’s units; RMSE is always at least as large as MAE and reacts more to large errors.
- MSE is the usual loss for training, because it can be differentiated.
- R² compares the model with always predicting the average: 1 is perfect, 0 is no better, below 0 is worse (possible on test data, never on the training data of a least-squares line).
- For one feature, training R² is the squared correlation r^2 .
- R² never falls on the training data when features are added; adjusted R² penalises each feature and so detects useless ones.

9. Sources

Built from

- CampusX, “Regression Metrics | MSE, MAE & RMSE | R2 Score & Adjusted R2 Score”, YouTube, <https://www.youtube.com/watch?v=Ti7c-Hz7GSM>
- StatQuest with Josh Starmer, “R-squared, Clearly Explained!!!”, YouTube, <https://www.youtube.com/watch?v=bMccdk8EdGo>. The link between R² and the squared correlation (section 6.3), and the idea of comparing the variation around the mean line with the variation around the fitted line (Figure 6); drawing the errors as squares is our own.

Other references

- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). *An Introduction to Statistical Learning* (ISLR), 2nd ed. Springer. §6.1.3 (adjusted R²).
- scikit-learn API reference, `sklearn.metrics.r2_score` (the score can be negative, because a model can be arbitrarily worse than predicting the mean). scikit-learn.org.
- Kaufman, S., Rosset, S., Perlich, C. and Stitelman, O. (2012). Leakage in Data Mining: Formulation, Detection, and Avoidance. *ACM Transactions on Knowledge Discovery from Data* 6(4).

10. Key terms

Term	Meaning
Feature	An input variable: one column of the data table
Target	The output we predict
Observation	One record: one row of the data table
Regression metric	A number that summarises how close a regression model's predictions are to the true values
Mean absolute error (MAE)	The average absolute difference between actual and predicted values
Mean squared error (MSE)	The average squared difference between actual and predicted values
Root mean squared error (RMSE)	The square root of MSE, in the target's units
R ² score (coefficient of determination)	1 minus the model's squared error divided by the squared error of always predicting the mean
Residual sum of squares	The total squared error of the model's predictions
Total sum of squares	The total squared error of always predicting the mean
Adjusted R ²	R ² with a penalty for the number of features
Target leakage	Building a feature from the answer itself, so the model sees information it would not have in real use